Statement of Purpose

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I am interested in how artificial intelligence can become more useful when it meets real-world constraints—limited computational resources, uncertain human behavior, and problems that originate outside computer science. My training in both finance and data science has naturally drawn me to research across AI systems, algorithmic decision-making, and interdisciplinary applications of AI. More broadly, I hope to develop methods and systems that allow people in other fields to use AI not only as a convenient tool, but as a way to investigate questions that would otherwise be difficult to study.

I did not initially know what research would look like. When GPT-4o was released, I became curious about why such a system could work at all: how a model processed a prompt, why attention mattered, and how an answer was generated from beginning to end. I began studying computer science independently through courses such as CS50, CS61B, CSAPP, and Advanced Natural Language Processing. Over time, however, learning existing material was no longer enough. I became more curious about how new knowledge was created in the first place, and whether ideas of my own could survive that process. With my background in finance and data science, I entered research through algorithmic game theory.

My first project studied revenue optimization in repeated second-price auctions. What initially seemed like a familiar auction setting became much less straightforward once current outcomes affected future opportunities. A bidder who won one round could receive an advantage in the next, so the incentive properties of a single auction no longer carried over automatically to repeated interactions. I contributed to the design and theoretical analysis of a threat-based boost mechanism and led much of its implementation and experimental evaluation, including revenue optimization with analytical gradients and projected gradient ascent. This project was my first experience moving between theory and computation, and it showed me how changing one assumption in a classical model could create an entirely different research problem.

I next joined a project on stable matching under probabilistic preferences. We were interested in whether LLMs could model the uncertainty in human decisions that classical Gale–Shapley matching does not capture. Rather than assuming that an agent always follows a fixed ranking, we allowed decisions such as remaining with a current partner or switching to another option to be probabilistic.

An early experiment forced me to rethink how we were evaluating the problem. Our method did not perform better than classical Gale–Shapley under the metric we were using. I went back through the experimental code and realized that we had been measuring match quality by summing the satisfaction scores that matched partners assigned to each other. The calculation was correct, but the metric was not measuring stability. Two people could both be satisfied with their current partners while still forming a pair that would prefer to leave those matches for each other.

That observation changed the direction of our evaluation. I proposed focusing on the probability of such deviations, which led us to develop *Expected Blocking Pairs* (EBP), a probabilistic extension of the classical blocking-pair notion. We then built an LLM-based preference framework grounded in quantal response theory and a Hybrid GS–LLM algorithm with probabilistic acceptance decisions. I led the implementation and experimental design across five LLMs and 21 real-world speed-dating groups, and wrote the methodology section of the paper. The resulting work, on which I am an equal first author, was accepted to ACL 2026. More importantly for me, this project was the first time an unexpected result led me not to immediately change the algorithm, but to question whether we were asking and measuring the right thing.

My current project, SPARE, began from a much more practical observation. While using online LLM services, I often experienced long response times and became curious about what happened when many requests competed for limited GPU resources. Reading work on LLM serving systems, including Sarathi-Serve, drew me toward the interaction between prefill, decoding, KV-cache allocation, and scheduling. I began asking whether a scheduler

could improve both first-token latency and the number of requests a system could sustain.

My first attempt was to learn scheduling decisions with reinforcement learning. In practice, this approach was difficult to make reliable. The quantities I cared about, such as mean latency and serving capacity, were only meaningful after many requests had been processed, so useful rewards arrived slowly. Training on GPUs was also expensive, and a policy learned under one workload or resource configuration did not necessarily remain effective when the environment changed. After spending substantial time on this direction, I began to suspect that learning a generic policy was avoiding the more important question: what structure of the serving problem could be exploited directly?

The breakthrough came from revisiting my earlier experiments. I noticed that reducing the token budget in an iteration could actually lower mean time to first token. I began tracing why. Unfinished prefills could consume computation and KV-cache space without producing an immediate output token; under memory pressure, allowing all of them to progress could interfere with requests that were closer to producing output.

This observation led me to explore joint choices of total token budget and per-request chunk size. I first introduced output protection so that ongoing decoding and prefills that could finish in the current iteration would not be unnecessarily delayed. I then encountered a more subtle issue: reducing scheduled tokens did not necessarily save memory, because KV cache is allocated in physical blocks. A scheduler could therefore delay computation without releasing any usable memory. I began estimating projected physical KV-block savings directly and used them, together with current memory pressure, unfinished prefill work, and recent arrivals, to determine when deferral was actually worthwhile. These ideas became SPARE, a KV-cache-aware online scheduler that selectively defers unfinished prefills while preserving immediate output opportunities.

Experimentation created a second bottleneck. Running every scheduler configuration directly on GPUs made it difficult to explore many workloads, models, and KV-cache settings. I therefore built a GPU-calibrated simulator aligned with vLLM V1, with interfaces designed to support different models, scheduling policies, and resource configurations. I reproduced the relevant scheduling and memory-allocation behavior and calibrated execution times from GPU measurements. Across four models, the framework reached 99.24

SPARE has also shaped the kind of research I want to pursue. I am increasingly interested in problems where the right solution depends on understanding the underlying system or domain rather than applying a method simply because it is powerful or fashionable. In this project, reasoning about physical KV-block allocation ultimately mattered more than continuing to improve a learned scheduler; in the matching project, reconsidering the definition of stability mattered more than tuning an algorithm against the wrong metric.

In graduate study, I hope to continue along two connected directions. First, I want to study efficient and adaptive AI systems, especially online scheduling and resource allocation under changing workloads, models, and hardware constraints. Second, I want to explore how AI can work with domain knowledge in areas such as economics, finance, and scientific research. I am particularly excited by the possibility that a finance researcher without extensive programming experience could use AI to test a sophisticated hypothesis, or that a scientist could use AI to identify a pattern that would otherwise remain hidden. My goal is not to force these problems into one technical framework, but to understand what each problem requires and develop methods that are both technically sound and genuinely useful.

What keeps me interested in research is that I do not know in advance where an investigation will lead. I enjoy learning ideas that are unfamiliar to me, tracing results that contradict my expectations, and turning vague questions into things that can be tested. I am willing to spend a long time on a problem even when the outcome is uncertain, as long as the question itself is worth understanding. A Ph.D. would give me the depth, time, and research environment to continue doing that work and to build systems and ideas that others can use to explore questions of their own.